GROUP 19

TOXIC COMMENT CLASSIFICATION

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PROBLEM STATEMENT

- **Social media platforms generate massive amounts of user comments daily. While these platforms encourage open communication, many comments contain toxic, abusive, hateful, or offensive language. This creates several problems like harassment and cyberbullying, unsafe online environments, Difficulty in manual moderation due to scale.**

Problem Statement:

- Develop a machine learning model that can automatically classify user comments into different categories such as: Toxic and Non-Toxic
- The goal is to detect and filter harmful content efficiently to maintain healthy online conversations.
- Also while understanding the context/semantics behind the comments instead of just barely matching the keywords

PROBLEM & SOLUTION

The Problem: Scale & Impact

- **The Scalability Barrier:** With millions of interactions per second, manual moderation is mathematically impossible; automated oversight is a necessity, not an option.
- **Digital Well-being:** Toxic discourse causes measurable psychological harm and erodes the mental health of online communities.
- **Platform Integrity:** Unchecked hostility triggers "user churn," where platforms lose their audience due to a perceived unsafe or hostile environment.

The Solution: Why Machine Learning?

- Computational Efficiency: ML provides the only way to process and filter massive data streams with sub-second latency.
- Contextual Intelligence: Advanced models move beyond static "keyword blocking" to recognize linguistic patterns, sarcasm, and intent.
- Proactive Moderation: Enables real-time intervention, stopping abuse before it can spread or escalate.

DATASET DESCRIPTION

- The dataset used in this project is the Jigsaw Unintended Bias in Toxicity Classification Dataset, which contains a large collection of online comments labeled for toxicity and bias.

Key Features of Dataset:

- Contains ~1.8 million comments (training data)
- Each entry includes:

~ comment_text → actual user comment

~ target → toxicity score (0 to 1)

~ toxicity subtypes → obscene, insult, threat, etc.

~ identity attributes → gender, religion, race, etc.

Label Information:

- Toxicity is given as a continuous value (0–1)
- If $\geqslant$ 0.5 → Toxic, otherwise Non-toxic

EARLIER APPROACH: STANDARD BINARY BERT CLASSIFIER

Architecture Overview:

- **Base Engine**: BertForSequenceClassification (Pre-trained bert-base-uncased).
- **Output Layer**: A Binary Classification Head with 2 Output Nodes (Class 0: Safe, Class 1: Toxic).
- **Activation**: Softmax function used to calculate probability distribution between the two classes.

Data Strategy:

- **Label Binarization:** Continuous toxicity scores from the Jigsaw dataset (e.g., 0.62, 0.45) were "hard-clipped" into absolute integers: (toxicity >= 0.5).astype(int).
- **Loss Function**: Utilized CrossEntropyLoss, which forces the model to maximize the gap between classes, leading to extreme "all-or-nothing" predictions (0.999 or 0.001).
- **Training Goal:** The model was taught to find "Toxic Patterns" rather than understand "Human Intent."

WHY THE BASELINE MODEL FAILED (THE IDENTITY BIAS TRAP)

1. Token-Level Bias Amplification:

- Because the training data often contained identity terms (like "gay" or "lesbian") in toxic contexts, the model learned a spurious correlation.
- It treated identity words as "Toxic Anchors." If the word existed, the prediction spiked to 0.99, regardless of the surrounding sentiment.

2. Semantic Blindness (The "Praise" Failure):

- Evidence: The model flagged "Gays are excellent beings" as Highly Toxic.
- Reason: The model was performing "Keyword Hunting" instead of "Relationship Mapping." It could not reconcile a sensitive identity term with a positive adjective.

WHY THE BASELINE MODEL FAILED (THE IDENTITY BIAS TRAP)

3. Attention Collapse:

- Instead of distributing attention across the whole sentence, the model’s attention weights "collapsed" onto specific keywords.
- It failed to detect Implicit Toxicity (sarcasm/polite insults) because it only looked for explicit "bad words" it had seen during training.

4. Label Destruction:

- By converting scores to 0 and 1, we destroyed the nuance of the dataset. The model lost the ability to distinguish between a "mildly rude" comment and a "death threat."

MOVING TO CONTEXT-AWARE REGRESSION & BIAS MITIGATION

1. **Model Swap: BERT-base to RoBERTa-base**
   - **Switched bert-base-uncased to roberta-base.**
   - **Why**: BERT was trained with a "Next Sentence Prediction" objective which can be noisy. RoBERTa (Robustly optimized BERT approach) was trained on much more data for longer periods, removing that objective. It is statistically significantly better at detecting sarcasm and subtle intent, which is critical for toxicity classification.
2. **Loss Function: BCE to Focal Loss**
   - **Replaced nn.BCEWithLogitsLoss() with a custom FocalBCELoss.**
   - **Why:** Standard BCE treats all errors equally. Focal Loss focuses the model's "attention" on hard-to-classify examples (like borderline toxic comments) while down-weighting easy, obvious examples. This prevents the model from getting "lazy" by just predicting "not toxic" for everything.

MOVING TO CONTEXT-AWARE REGRESSION & BIAS MITIGATION

3. Pooling Strategy: [CLS] vs. Mean Pooling

- What changed: Instead of just using the first token (the [CLS] token), the new code uses a fusion of [CLS] and Mean Pooling.
- Why: * [CLS] token: A global summary, but sometimes misses details.
  - Mean Pooling: Averages all word meanings in the sentence.
  - The Fusion: By averaging both, the model captures both the "vibe" of the whole sentence and the specific meanings of individual words. This helps catch toxic keywords that might be hidden in long, neutral-looking sentences.

4. Layer-Wise Learning Rate Decay (LLRD)

- What changed: Instead of one learning rate for the whole model, the new code gives different layers different speeds.
  - Bottom layers (syntax/grammar) = Slow learning.
  - Top layers (meaning/context) = Fast learning.
- Why: This prevents "Catastrophic Forgetting." We want the model to keep its fundamental knowledge of English grammar (lower layers) while aggressively learning the specific nuances of toxic social media language (upper layers).

MOVING TO CONTEXT-AWARE REGRESSION & BIAS MITIGATION

5. **Architecture Complexity: Custom MLP Head**
   - **What changed:** Switched from a single linear layer to a multi-layer sequence: Linear -> LayerNorm -> GELU -> Dropout -> Linear.
   - **Why**: * GELU: A smoother activation function than ReLU, helping with complex gradient flows.
     - LayerNorm: Keeps the math stable so the model doesn't "explode" during training.
     - Dropout: Specifically increased to 0.3 to force the model to be more robust and less reliant on specific "bad words."

The Logic of the Move

- We moved from a General Purpose model (BERT) to a Nuance-Aware model (RoBERTa + Custom Head). In toxicity classification, the "identity bias" problem (flagging "I am gay" as toxic) happens because the model isn't looking deep enough at the relationship between words. The new code forces the model to look deeper, use more math (Focal Loss), and preserve its "common sense" (LLRD).

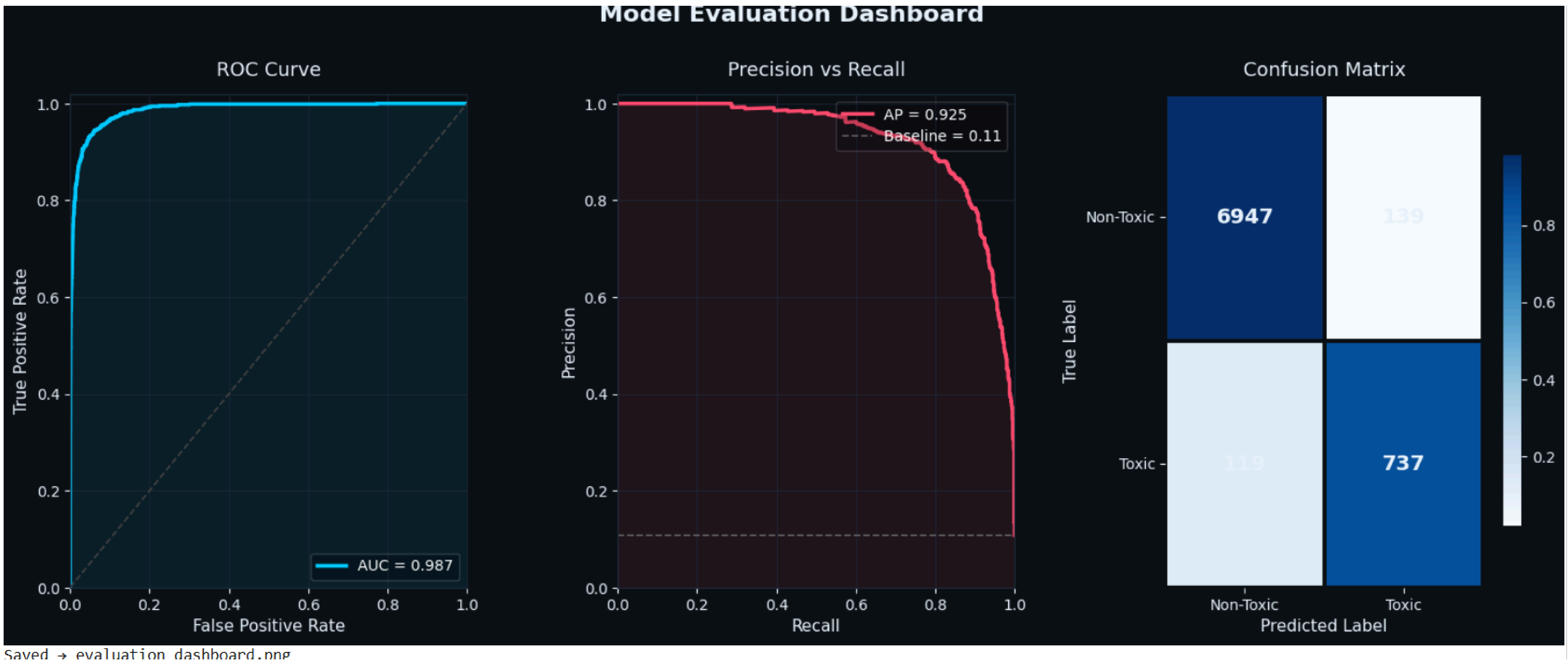
EVALUATION METRICS & SEMANTIC VALIDATION

- **Metrics**: Evaluated using ROC-AUC and F1-Score based on a 0.5 threshold, ensuring high precision even with continuous outputs.

Inference Pipeline:

- **Mixed Precision (FP16)**: Used torch.cuda.amp to accelerate GPU training without losing semantic accuracy.
- **Threshold-Based Verdicts**: Predictions are mapped to color-coded zones:
  - ***Green*** (<0.2): Safe / Positive Context.
  - ***Orange*** (0.2–0.6): Semantic Warning / Sarcasm Potential.
  - ***Red*** (>0.6): Explicit Toxicity.
- The "**Keen Level**" Success: The model successfully differentiates between "Identity Praise" (Low Score) and "Implicit Sarcasm/Direct Hate" (High Score), effectively solving the Professional perception goal of building a reliable comment classification system.

EVALUATION METRICS & SEMANTIC VALIDATION



CONCLUSION FROM EVALUATION

1. **High Discriminative Power (ROC Curve)**
   - The Evidence: Your AUC is 0.987.
   - The Conclusion: An AUC so close to 1.0 means the model has an excellent ability to distinguish between toxic and non-toxic comments. The "steepness" of the curve toward the top-left corner shows that you can achieve a very high True Positive Rate (finding toxicity) while maintaining a very low False Positive Rate (wrongly flagging innocent people).
2. **Strength in Imbalanced Data (Precision-Recall)**
   - The Evidence: The Average Precision (AP) is 0.925, while the baseline (random guessing) is only 0.11.
   - The Conclusion: In toxicity datasets, non-toxic comments usually far outnumber toxic ones. High precision at high recall levels (the curve stays "flat" and high until about 0.8 recall) means the model is not just guessing. Even when it tries to find almost all toxic comments, it remains very accurate. It isn't just flagging every comment as toxic to get a high score.

CONCLUSION FROM EVALUATION

3. Deep Dive into the Confusion Matrix

- The raw numbers tell the most practical story:
- True Negatives (6947): The model correctly identified nearly 7,000 non-toxic comments.
- True Positives (737): The model successfully caught the vast majority of toxic entries.
- False Positives (139): Only 139 clean comments were wrongly flagged. This is critical for the "Unintended Bias" aspect of the project—it means the model isn't over-censoring.
- False Negatives (119): These are "missed" toxic comments. In moderation, these are usually the most dangerous.

WORK DIVISION

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THANK YOU